MTHM044 Projects: 2026/27

Below you will find details of the academic members of the Statistics and Data Science group in the Department of Mathematics and Statistics. Further details can be found here. Please consider each member to be potential supervisor of your project. To choose a project, first consider *your* research interests, and then consider which supervisor might be best suited to supervise you. Please contact and meet with (in person or online) any potential supervisor. Once you've chosen a supervisor and project title, please pass the information to the MTHM044 module leader, Ben Youngman. Please also contact Ben if you have any queries about the module, or seek help finding a supervisor or consolidating your research interests.

- **Peter Challenor:** uncertainty quantification, such as statistical analysis of complex numerical models (such as those used to simulate climate and in healthcare). The estimation of the numbers visiting green spaces, such as nature reserves.
- **Theo Economou:** climate change impacts, environmental health, hierarchical models
- **Christopher Ferro:** statistical forecasting.
- **Paul Hewson:** survey methods, specifically model-based post-stratification and spatial models (applied to accessibility and safety issues in road transport), official statistics.
- **Mark Kelson:** clinical trials, medical statistics, causal inference and data science with applications to physical activity and mental health.
- **Hossein Mohammadi:** Design and analysis of computer experiments which includes adaptive/sequential sampling, sensitivity and uncertainty analysis; Statistical modelling based on Gaussian process emulators.
- **Chaitra Nagaraja:** time series, socioeconomic and macroeconomic measurement, official statistics, survey sampling, history of statistics.
- **Tarje Nissen-Meyer:** geophysics, seismology, time series, inverse problems machine learning, computational modelling, wildlife monitoring, soil monitoring
- **Oscar Rodriguez de Rivera Ortega:** Ecological Statistics, Species Distribution Models, Spatial and Spatio-Temporal modelling
- **James Salter:** Emulation and calibration of environmental models, uncertainty in ML/AI models, model discrepancy, Bayesian inversion, bias correction.
- **Stefan Siegert:** spatial/timeseries/spatiotemporal modelling, statistical forecasting, applications to environmental sciences, combining computer simulation models with historical observations, assessing forecast model performance.
- **David Stephenson:** statistical modelling of weather and climate processes, with applications including improving our ability to forecast and reduce risk.
- **Victoria Volodina:** uncertainty quantification, emulation of complex computer models, graphical models for decision-support, often in application to climate modelling, energy systems planning and physiological monitoring in critically ill patients.
- **Ben Youngman:** extreme value modelling, spatial statistics, computational statistics and interpretable AI, often with applications to extreme weather.

Forecast updating

Module pre- or co-requisites: MTH1004, MTH2006, and preferably some third-year statistics modules too
Supervisor: Chris Ferro

Forecasters should update their forecasts when new information arrives, but the forecasters should neither over-react nor under-react to the new information. We have been developing statistical methods for assessing whether forecasters update their forecasts in appropriate ways. If they are updated appropriately then the forecasts are called efficient (or rational). If the forecasts are inefficient then the updates may be predictable, which can make it possible to create better forecasts.

In this project, you will learn about the theory of assessing forecast updates and then investigate whether real forecasts show signs of being inefficient. Forecast from various sectors may be used, e.g. weather forecasts, economic forecasts or prediction markets. The project requires good experience of probability, statistics and computing.

Forecast optimism

Module pre- or co-requisites: MTH1004, MTH2006, and preferably some third-year statistics modules too
Supervisor: Chris Ferro

When we create a forecasting model, it's useful to know how well it will perform. If we test its performance using the same data that we use to select and fit the model, however, then our assessment will be optimistic. Such optimism can be reduced with various forms of cross-validation and bootstrapping. This project will compare the effectiveness of different ways of reducing optimism.

In this project, you will learn the key principles of creating and applying statistical forecasting models. The project may focus on regression or time-series models suitable for forecasting in domains such as environmental science or finance and economics. The project requires good experience of probability, statistics and computing.

Calibrating Complex Numerical Models (ODE/PDE/SDE) which have multiple solutions

Module pre- or co-requisites: None (MTHM033 would help, but it is not essential)
Supervisor(s): Peter Challenor, Marc Goodfellow, Victoria Volodina

Often we use sets of ordinary partial or stochastic differential equations (ODE/PDE/SDE) to model real world phenomena (the example we will use in project is a set of ODEs to model brain activity in neo-natal babies). These models depend on unknown model inputs which need to be estimated from data on the model outputs, this is known as calibrating the model or inverse modelling. We also need to take into account the fact that the model isn't perfect. There is a discrepancy between the model and reality.

The method we use for such problems is called History Matching. We numerically solve the differential equations in a small number of carefully selected places. We then build what is known as an emulator for the model. An emulator is a fast, stochastic approximation to the full set of differential equations, with known uncertainty. (Numerically solving thedifferential equations is computationally expensive). We can now compare the emulated solution at any set of input values with the data, calculating what is known as implausibility. We remove those points that are implausible – if we remove all the implausible points what remains must include the 'best' point. Initially our emulator is a poor approximation and the reduction in NROY (not ruled out yet) space may be little. But we keep doing more numerical solutions, and building better emulators in NROY, in waves and after a few waves we soon reduce NROY to a small percentage of the original input space.

We know from exhaustive enumeration across input space of our brain model that there are two solutions. There is a bifurcation in the ODE. When we carry out History Matching we find the one solution, but we do not find both. The project is concerned with modifying the History Matching procedure to work reliably with multiple solutions. We have ideas of possible solutions, but we will encourage the student the develop their own. The results should be publishable. Coding will be done in R and use the HMER package on CRAN to a large extent. The project should suit a student who is interested in modern statistical methods at the edge of machine learning, AI and applied maths.

Parameter Estimation in mathematical model from ECG data using functional history matching

Module pre- or co-requisites: MTH2006
Supervisor(s): Victoria Volodina, James Salter and Sandip George (University of Aberdeen)

Mathematical models are important tools for gaining insight into complex processes across biological systems and healthcare. To make these models actionable and relevant in practice, it is crucial to calibrate them using real-world observations while accounting for their limitations. In this project, we will consider a delay ordinary differential equation model that outputs synthetic electrocardiogram (ECG) signals. The internal parameters of the model are designed to reflect key physiological mechanisms that influence ECG rhythm. To facilitate the use of these tools for diagnostic purposes, developing efficient and fast methods for learning model parameters by comparing clinical ECG data with model output is a crucial step in this process.

We propose to adopt approaches from inverse uncertainty quantification (UQ), history matching (HM), to perform model parameter estimation from patients' ECG data. Traditionally, history matching is performed by iteratively ruling out regions of input parameter space that are considered unlikely to produce a model output consistent with real-world observation, while explicitly accounting for model discrepancy and observational uncertainty. Since ECG data is naturally considered as functional data, in this project, we are interested in applying the functional history matching method (FHM). This research topic raises several important and practical questions that the student can explore, such as how to construct implausibility measures for functional data, i.e., the appropriate choice of a distance function and the implausibility threshold, as well as how to specify model discrepancy (error) for functional data in the clinical context.

The student on this project will take a leading role in developing and applying functional history matching to estimate model parameters from ECG data using pre-processed data from a small cohort of patients with no co-morbidities.

Uncertainty and errors in volcanic ash forecasting

Module pre- or co-requisites: MTH3041 or MTHM047 would be beneficial
Supervisor(s): James Salter, Helen Webster (Met Office)

When a volcano erupts, the Met Office is required to produce forecasts of how the ash cloud is going to evolve in order to produce advisories for health and aviation. To do so, their operational system ('InTEM') combines prior estimates of the height/time emissions profile of the volcano with satellite observations of the ash cloud, and uses the NAME atmospheric dispersion model to simulate the future evolution of the cloud given meteorological forecasts.

The InTEM system uses Bayesian inversion and Non Negative Least Squares to find an optimal estimate of the emissions, and uses this estimate when producing forecasts of the ash cloud. Recent work has extended this system to provide uncertainty on the height/time emissions profile and use this to produce probabilistic forecasts, but other sources of uncertainty are either not directly accounted for (in the dispersion model, in meteorology), or are accounted for by imperfect assumptions (independent Normal assumption on satellite observation error).

This project will explore further improvement of and better quantification of uncertainty within the InTEM system, with the end goal of improving the quality of probabilistic forecasts provided. The project could consider any combination of a) alternative priors, b) alternative observation error assumptions, c) uncertainty in the dispersion model and driving meteorology d) model discrepancy, the difference between model-world and reality.

Bias correcting marine chlorophyll forecasts

Module pre- or co-requisites: MTH3045 may be beneficial
Supervisor(s): James Salter, Met Office supervisor TBC

Plymouth Marine Lab and the Met Office provide daily forecasts of the concentration of phytoplankton chlorophyll in the seas around the UK, given by assimilating observations into a biogeochemical model. Due to imperfections in the data and in the underlying process-based model, these forecasts contain biases, both spatially and at certain times of year, in particular tending to over-predict chlorophyll on days 3-6 of the forecasts when plankton are growing strongly, whilst in other cases extreme events are missed entirely.

The forecasts can be brought closer to the observations via 'bias correction'. Past projects have demonstrated improvements at a small set of locations via methods like Gaussian Processes and XGBoost, but challenges remain in rolling this out more widely across the full spatial field, and in identifying the causes of different biases and drivers of extreme events.

This project has several interesting directions, with the main goal of attempting to bias correct the full spatial field around the UK. This could be achieved by comparing a few standard statistical/ML methods, or by exploring in more detail a particular state-of-the-art ML/AI methodology (e.g., deep Gaussian processes, DeepKriging, neural operators).

Statistical AI approaches for quantifying the impact of climate change on human health

Module pre- or co-requisites: MTH2006 or basic statistical knowledge
Supervisor: Theo Economou

One of the biggest concerns relating to global warming is the negative impact on human health. The frequency of extreme hazards such as heatwaves, coldwaves, droughts and air pollution is increasing but the impact on human health and therefore health resources is not well understood. This project will investigate the use of modern AI models for analysing health and environmental exposure data for flexible modelling of the health effects and for the derivation of interpretable and actionable quantities. Conventional modelling methodologies in environmental epidemiology lack capacity in terms of scalability and flexibility, therefore the goal of this project is to investigate the use of modern methods in machine learning and AI as ways to extend or complement existing methods broadly based on classical regression.

Quantifying the health burden of a warming UK: heatwaves, wildfires and statistical AI methods in epidemiology

Module pre- or co-requisites: MTH2006 or basic statistical knowledge
Supervisor: Theo Economou

The project is motivated by the lack of available/reliable estimates of how climate change is affecting health in the UK. In particular, the idea is to use data on health outcomes (mortality), meteorological variables (temperature, humidity, etc.) and atmospheric information (air pollution due to wildfires), in conjunction with modern data modelling methods (statistical AI) to quantify the disease burden from events such as the heatwaves and wildfires of the summer of 2026.

Predicting and understanding athletics records using extreme value theory

Module pre- or co-requisites: MTH2006, and one or more of MTH3028, MTH3041, MTH3045 may be beneficial
Supervisor: Ben Youngman

Extreme values can be defined as the very big, or the very small. And in athletics, it is often the very small that we want, such as the fastest time in a race. There is a lot that extreme value theory [1] can help us explore: What is the fastest time that might be humanly possible [2]? Is a record consistent with the past [3]? Or has a technology brought a step-change in times possible?

This project will initially develop extreme value models for athletic records, and then will explore various models for nonstationarity for answering questions such as those raised above in the light of many more years of data. The project requires good experience of data analysis, statistical modelling and computing using R.

(1) Coles, S. G. (2001) An Introduction to Statistical Modelling of Extreme Values. Springer.
(2) Einmahl, J. H. J., & Magnus, J. R. (2008). Records in Athletics through Extreme-Value Theory. Journal of the American Statistical Association, 103(484), 1382–1391.
(3) Robinson, Michael E., and Jonathan A. Tawn. “Statistics for Exceptional Athletics Records.” Journal of the Royal Statistical Society. Series C (Applied Statistics) 44, no. 4 (1995): 499–511. https://doi.org/10.2307/2986141.

Short-term prediction of extreme weather using extreme value theory

Module pre- or co-requisites: MTH2006, and one or more of MTH3028, MTH3041, MTH3045 may be beneficial
Supervisor: Ben Youngman

Extreme value theory is a branch of statistics where we focus on extreme data, and often want to extrapolate to the even more extreme. For example, when building a sea defence, we might have a century of tidal data, but want to build our defence to withstand the kind of inundation that's expected only once in 10,000 years. To date extreme value analyses have typically focussed such probabilistic estimation to give long-term protection.

However, we might be interested in a more short-term view of extremes. For example, given the last week's rainfall, and perhaps other information such as soil saturation, what's the probability that tomorrow's rainfall exceeds 100mm? Such estimates could be improved by incorporating weather forecast data. Short-term estimates like these can serve as a warning as to whether we might expect flooding, and might want to take immediate action to guard against its effects. This project will develop models of extremes related to extreme weather given preceding weather conditions, and will explore the value of drawing in weather forecast data. The project is likely to involve modelling fairly large amounts of data, programming in R, and interpreting results in the context of extreme weather prediction.